

Practice Questions on Convolution Neural Networks

1. Consider a 4 X 4 array of numbers which after convolution with a kernel of size $w \times h$, stride s_w, s_h , and unknown padding gets transformed to the array alongside.

2	5	10	17
5	8	13	20
10	13	18	25
17	20	25	32

0	0	0
0	0	0
0	0	0

- (a) Guess a non-zero kernel matrix and values of stride and padding that could have transformed the left array to the right array. ..3 One possible kernel matrix is:

1	-1
-1	1

Others may also be possible. No padding. Stride is 1, 1

- (b) Suppose we now try to find the kernel parameters using gradient descent with initial kernel entries to be all ones and loss function as the square difference of kernel-computed entries and true value (which is 0 in this case). Write the gradients compactly using double summation using e_{ij} to denote the original matrix entries and $w_{11}, w_{21}, w_{12}, w_{22}$ to denote the parameters of the kernel. Your answer will be of the form: $\sum_{i=1}^3 \sum_{j=1}^3$ some function of parameters and entries ..2 For k_{ab} for $a = 1, 2, b = 1, 2$

$$\sum_{i=1}^3 \sum_{j=1}^3 2(w_{11}e_{ij} + w_{12}e_{i,j+1} + w_{21}e_{i+1,j} + w_{22}e_{i+1,j+1})e_{i+a-1,j+b-1}$$

- (c) Inspecting the gradients above comment on the trajectory of values that the kernel parameters will take using gradient descent with a reasonable learning rate (say 0.1).

..3 We see that the gradients are all positive

to start with. Also the absolute value of the gradient component wrt w_{22} is always the largest and w_{11} is always the smallest with other two values in-between. This implies that gradient descent will find a solution where w_{22} will be the smallest and w_{11} will be the largest and the other two will be in-between.

2. Here we need to design a 2×2 kernel W with a bias parameter b that is applied on an image to first compute a convolution,

$$(W * X)_{i,j} = ReLU(W_{0,0}X_{i,j} + W_{0,1}X_{i,j+1} + W_{1,0}X_{i+1,j} + W_{1,1}X_{i+1,j+1} + b) \quad (1)$$

then followed by average pooling over all positions.

- (a) The W, b should be designed so that it can separate out image of a vertical 1 from image of digit 0. Assume the thickness of the outline of the digit is a single pixel. Note the digit can be placed anywhere on the image. ..2 $w_{0,0} = 1, w_{1,0} = 1$ rest are zero.
- (b) What alternative design choice do you recommend if your goal is to perform three way classification to distinguish among horizontal, vertical, and 45% slant lines. ..3 Add two more channels and design a suitable kernel to identify horizontal and slanted lines respectively..